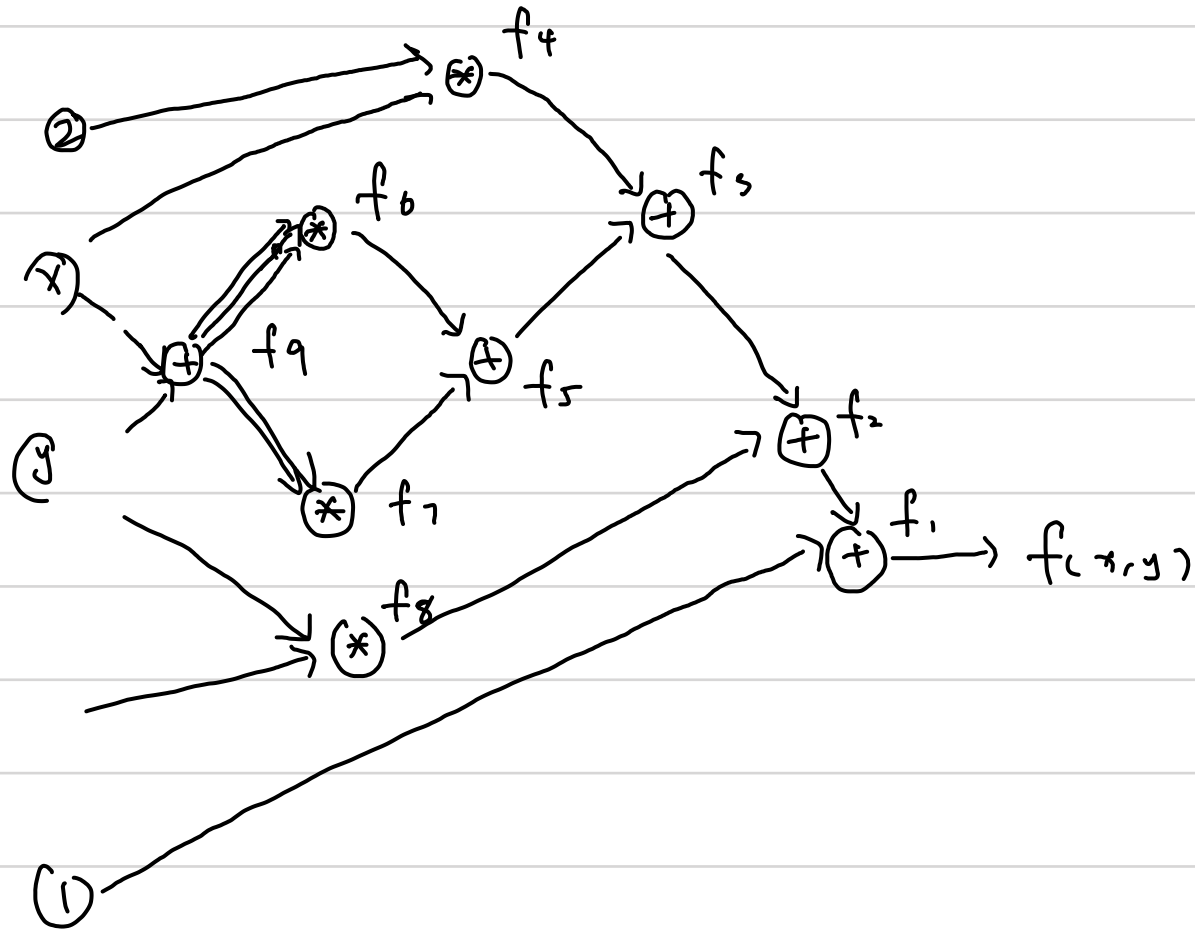
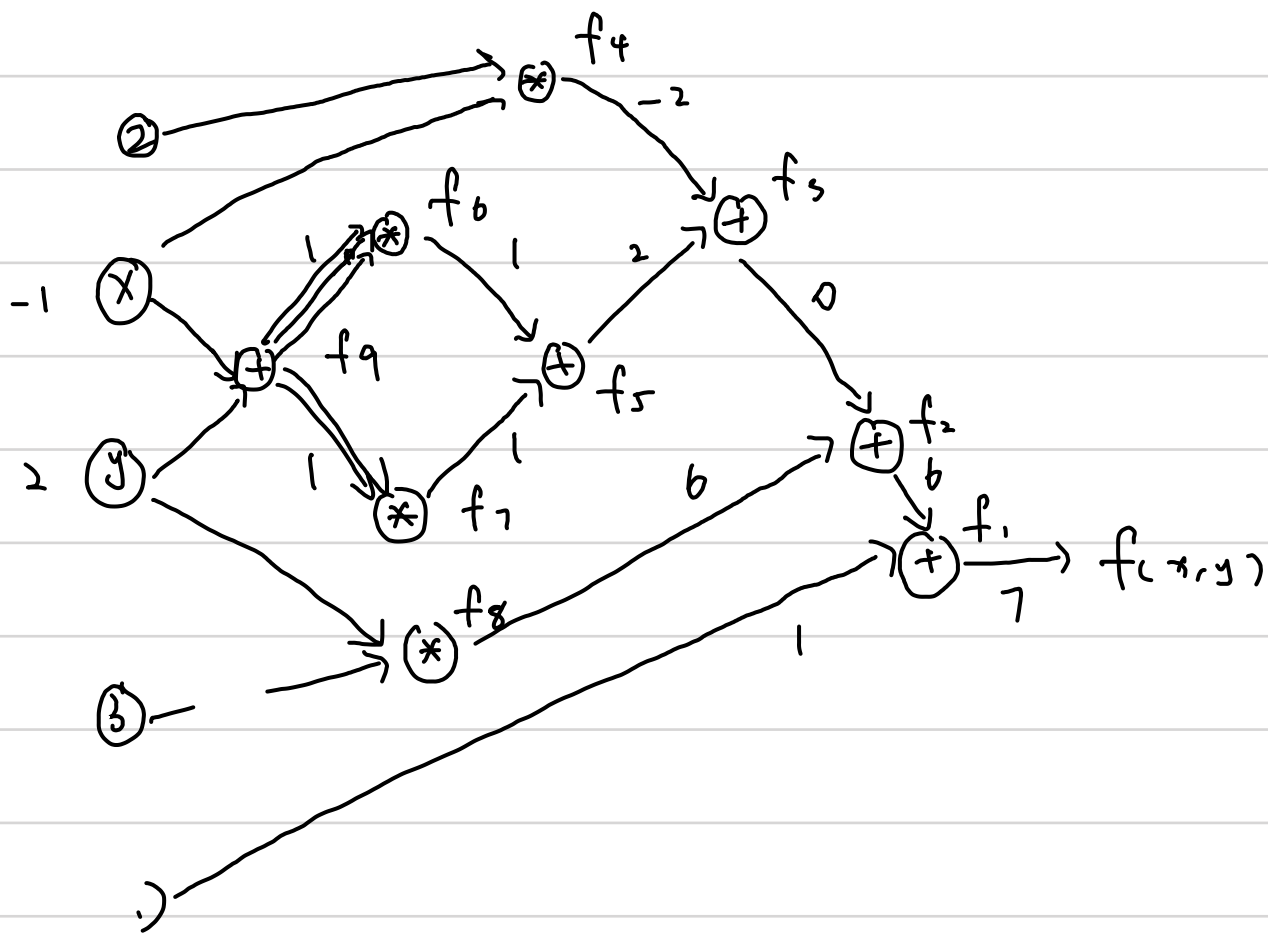
Question 1

hw1

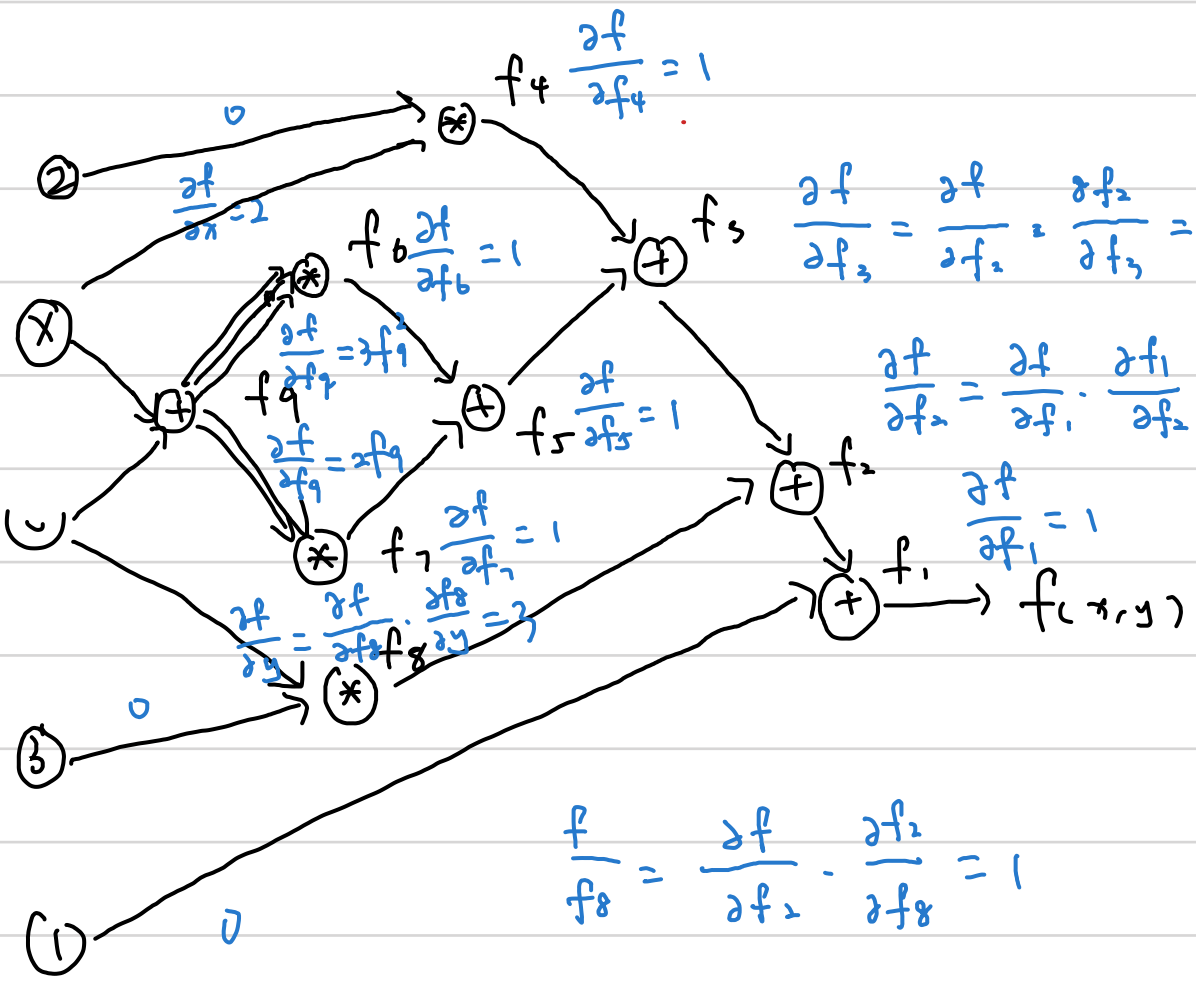
a)



b)



c)



$$\frac{\partial f}{\partial f_3}=\frac{\partial f}{\partial f_2}\cdot\frac{\partial f_2}{\partial f_3}=\frac{\partial(f_3+f_8)}{\partial f_3}=1$$

$$\frac{\partial f}{\partial f_2}=\frac{\partial f}{\partial f_1}\cdot\frac{\partial f_1}{\partial f_2}=\frac{\partial(f_2+1)}{\partial f_2}=1$$

$$\frac{\partial f}{\partial f_1}=1$$

$$\frac{f}{f_8}=\frac{\partial f}{\partial f_2}\cdot\frac{\partial f_2}{\partial f_8}=1$$

$$\frac{\partial f}{\partial x}=3f_9^2\cdot\frac{\partial f_9}{\partial f_x}+2f_9\cdot\frac{\partial f_9}{\partial f_x}+2=3(x+y)^2+2(x+y)+2$$

$=7$

$$\frac{\partial f}{\partial y}=3f_9^2\cdot\frac{\partial f_9}{\partial y}+2f_9\cdot\frac{\partial f_9}{\partial y}+3=$$

$$3(x+y)^2+2(x+y)+3$$

$=8$